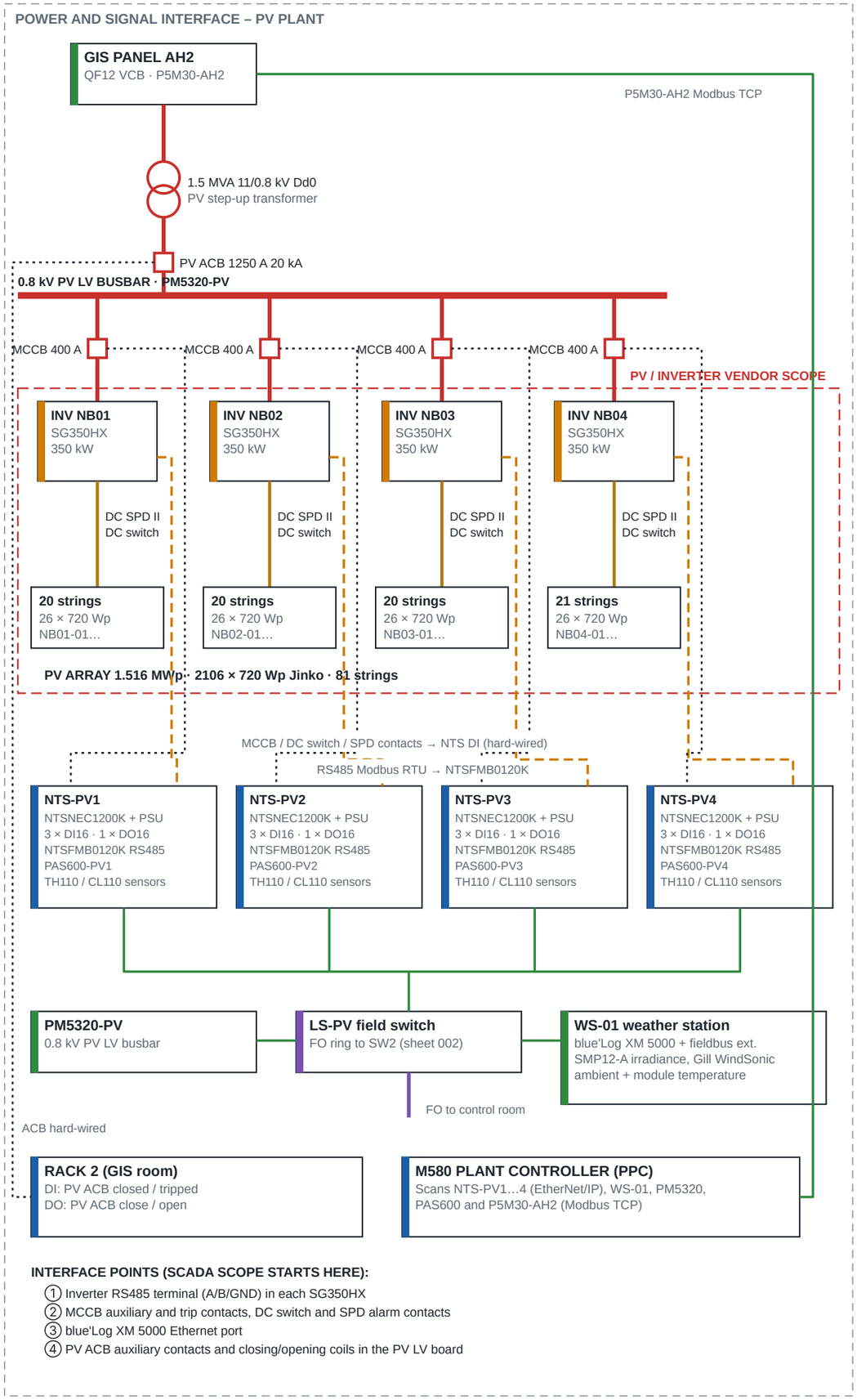


TABLE A – SIGNAL EXCHANGE (PER INVERTER UNLESS STATED)

Dir.	Signal	Type	Source	Rate
IN	AC P, Q, U, I, f; DC P, U	AI comm	SG350HX	2 s
IN	Energy today / total	AI comm	SG350HX	60 s
IN	Internal temp., insulation resistance	AI comm	SG350HX	10 s
IN	Operating state, fault code	AI comm	SG350HX	2 s
IN	String currents (20 / 21 per inverter)	AI comm	SG350HX	10 s
OUT	Active power limit (% of 350 kW)	AO comm	PPC	on change
OUT	Reactive power / PF setpoint	AO comm	PPC	on change
OUT	Start / stop	AO comm	Operator	command
IN	MCCB 400 A closed / tripped	DI	NTS-PVn	20 ms
IN	DC switch closed, AC/DC SPD failed	DI	NTS-PVn	20 ms
IN	Enclosure temperature / humidity	AI comm	PAS600-PVn	60 s
IN	POA irradiance, Tamb, Tmod, wind	AI comm	WS-01 (plant)	5 s
IN	PV LV P, Q, U, I, PF, energy, THD	AI comm	PM5320-PV	1 s
IN	PV ACB closed / tripped	DI	Rack 2	20 ms
OUT	PV ACB close / open	DO	Rack 2	command
IN	AH2 protection, measurement, events	DI/AI comm	P5M30-AH2	1 s

Signal count (I/O list, area PV): 18 DI · 1 DO · 155 AI comm · 12 AO comm · plus weather station 7 and AH2 relay points.

TABLE B – CONTROL FUNCTIONS AND BEHAVIOUR

Function	Description
Curtailment	PPC writes P limit to all inverters when BESS reaches charge limit or SOC ≥ 97 %.
	Ramp rate 10 %/s (TBC with BESS vendor).
Reactive power	Inverters at PF 1.0 by default; Q setpoint available to PPC.
Close permissive AH2	Bus live (BESS or DG forming the grid). PV cannot energise the bus.
Loss of grid former	Inverters trip on their own V / f protection (anti-islanding).
Comms loss PLC ↔ inverter	Inverter comms watchdog 30 s → fallback P limit (vendor setting, TBC).
	SCADA raises P2 alarm, values shown with bad quality.
String monitoring	String current vs. array average: warn at −10 %, alarm at −30 %.
Performance ratio	Calculated in SCADA from PV energy and WS-01 irradiation.

TABLE C – INTERFACE RESPONSIBILITY

Item	PV / inverter vendor	SCADA contractor
Modbus register map SG350HX	Supply	Implement
RS485 cable inverter → NTS	Terminate at inverter	Supply, lay, test
Inverter Modbus ID / baud setting	Set	Verify
Auxiliary contacts MCCB / SPD	Wire to terminal strip	Wire to NTS
Weather station sensors & logger	Supply, install	Integrate (Modbus TCP)
Curtailment test at SAT	Witness	Perform

LEGEND

- Power circuit
- DC circuit
- RS485 – Modbus RTU
- EtherNet/IP (M580 RIO / DIO)
- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- Hard-wired I/O (DI / DO)
- Vendor scope boundary

NOTES

1. Ratings from Overall SLD and inverter wiring diagram BFE/PR/26/01.
2. One NTS island and one PAS600 per PV area as per the offer (Inverter & DC combiner 1–4).
3. The BOM line PVSCL60E (Schneider string inverter) does not apply: the plant uses Sungrow SG350HX. Remove from BOM.
4. Modbus register map of SG350HX to be obtained from Sungrow before FAT.
5. Signal names follow the I/O list (area PV, tags VS_PV_INV_NB0x_…).

REV	DESCRIPTION	DATE
A	Draft for internal review	16-OCT-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
PV INTERFACE
Signal exchange, control and scope boundary

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
DESIGNED		SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN
DRW NO.	**BFE/PR/26/01-SCADA-003**	**REV A**